

Casey Mathieson

Systems Engineer | Precision Automation & Metrology

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Professional Summary

Electro-mechanical systems engineer with 11+ years of experience designing, building, commissioning, and improving automation equipment, manufacturing processes, and precision-metrology systems. Brings end-to-end ownership from root-cause analysis and concept development through mechanical/electrical design, fabrication, test, deployment, and durable process documentation. Combines hands-on manufacturing knowledge with strong mechanical design, controls, calibration, and cross-functional delivery skills.

Technical Expertise

Systems engineering & metrology: Precision metrology, laser calibration, sensors/instrumentation, motion and robotic systems, pneumatics, thermal control, test-method development, and root-cause analysis. **Mechanical design & manufacturing:** SolidWorks (CSWP), PDM, GD&T, complex assemblies, FEA/simulation, fixtures/nests, DFM/DFA, CNC/CAM/G-code, and fabrication. **Electrical, controls & functional safety:** Ground-up AC/DC panel design; schematic review/redlines; motor control/VFDs; PLCs; and machine-safety design, programming, and validation. **Commissioning, quality & process:** System commissioning, FAT/SAT, calibration, verification and validation, PFMEA, SOPs/work instructions, risk/dependency mapping, and process improvement. **AI-enabled engineering tooling & data workflows:** Agent-assisted tool development and validation; technical research, knowledge/file discovery, workflow automation, Python/MATLAB/TypeScript tools, and structured data. **Delivery & technical coordination:** Requirements definition, estimates/proposals, cross-functional coordination, customer and field support, technician-ready documentation, and knowledge transfer.

Professional Experience

ZeroTouch Metrology / DWFritz Automation — Wilsonville, OR

2020–Present

Systems Engineer, ZeroTouch Segment · 2024–Present

- Revived a stalled precision-stage qualification process for a ~\$20K-build/\$40K-buy component used across roughly 12 systems annually, progressing from legacy-method audit and control-system migration through automated micrometer-scale analysis, validation testing, and controlled SOPs.
- Owned redesign, validation, and deployment of full-machine Golden Calibration, reducing the workflow from roughly 14 elapsed hours with two people to 2–3 hours with one person through automated data import, pass/fail decisions, and controlled exports; trained two field-service engineers.
- Supported 18 precision-metrology system builds and led end-to-end commissioning on 6, coordinating mechanical, electrical, and software integration, calibration, verification, and deployment; owned Golden Calibration on 9.
- Overhauled component-level laser calibration from mechanical design and MATLAB/Python tooling through SOPs, execution on ~60 laser heads, and training five colleagues.

Mechanical Engineer · 2020–2024

- Modernized legacy machine designs and developed mechanical solutions for conveyance, packaging, part handling, and robotic end-of-arm tooling.
- Designed precision fixtures, nests, and adjustment stages.
- Improved calibration methods, DFM, CAD/BOM quality, and technician-ready documentation.
- Developed sensor- and dewpoint-based thermal-control logic to prevent condensation damage and enable safe operating responses.

Evergreen Advancements — Oregon

2023–Present

Founder / Product Systems Engineer · [cascadia.studio](#) · [type-hype.com](#)

- Independently architected and built a multi-brand product-operations platform using agentic workflows, integrating data models, asset preparation, pricing, marketplace publishing, analytics, and lifecycle controls.
- Built Cascadia Digital's Next.js/Firebase storefront with catalog search, checkout/download fulfillment, telemetry, responsive end-to-end testing, and fail-closed deployment.
- Built TypeHype, a zero-auth semantic font-discovery product with vision enrichment, pgvector retrieval and reranking, observability, cost controls, and canary validation across 250+ fonts and 500+ prewarmed discovery routes.

Sig Sauer Electro-Optics — Wilsonville, OR

Oct 2019–Feb 2020

Manufacturing Engineer

- Designed and integrated automated equipment and tooling for optical-electronics assembly.
- Improved reliability, throughput, and safety through equipment troubleshooting, six-axis robot programming, and PLC changes.

Production Robotics Inc. — San Leandro, CA

May 2015–Jul 2019

Senior Automation Engineer · Nov 2016–Jul 2019

Mechanical Design Engineer · May 2015–Nov 2016

- Designed and delivered custom manufacturing and assembly automation programs ranging from \$30K to \$1.5M.
- Performed design reviews, PFMEAs, FAT/SAT, and high-risk prototype/process testing across simultaneous projects.
- Designed mechanical assemblies and ground-up AC/DC electrical panels, including motor control, VFDs, and NFPA/NEC/NEMA-aligned schematics.
- Developed estimates, proposals, schedules, and technical delivery plans for robotic assembly and material-handling systems.

West Coast Surgical LLC — Half Moon Bay, CA

Oct 2014–May 2015

CNC Machinist

- Produced high-mix, low-volume medical-device work.
- Optimized milling setups, G-code, and workholding fixtures to meet tolerance requirements.

Shaffer's Offroad — Alameda, CA

Jun 2014–Oct 2014

CAD Designer / Fabricator

- Designed and manufactured custom vehicle components using SolidWorks, CNC plasma, press-brake, milling, and MIG-welding processes.

NASA Ames Research Center — Moffett Field, CA

Jun 2013–Oct 2014

Engineering Intern, Volunteer

- Contributed to R&D for a cable-actuated tensegrity rolling robot and led a senior-project team that prototyped a modular motorized cable-actuation system.

Education & Credentials

B.S., Mechanical Engineering, Mechatronics Focus — San Francisco State University, 2015

A.A., Economics; A.A., Marketing, Graphic Design Focus — Santa Barbara City College, 2011

Certified SOLIDWORKS Professional — Mechanical Design — Certificate C-QNHPJK9BLV, 2019